BCSE497J – Project-I

FEDERATED LEARNING FOR PRIVACY-PRESERVING MEDICAL DIAGNOSIS

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ABSTRACT

Clinical deep learning is bottlenecked not by model capacity but by data access. Individual hospitals hold datasets that are small, demographically narrow, and acquisition-biased, while regulatory frameworks such as HIPAA, the GDPR, and India's Digital Personal Data Protection Act make raw-data centralisation infeasible. Federated learning (FL) addresses this by keeping data in place and aggregating only model updates — but FL alone is not a privacy guarantee, since shared gradients are invertible via reconstruction and membership-inference attacks, and the open participation model exposes the global model to poisoning and Byzantine faults from compromised or misconfigured clients.

This project proposes a hierarchical client–server federated architecture for medical diagnosis, in which each leaf node (hospital) performs local training and a centre node performs secured aggregation. Two defence layers are embedded directly into the training loop: client-level differential privacy via per-sample gradient clipping and calibrated Gaussian noise under a Rényi-DP accountant, yielding a stated privacy budget (ε, δ); and Byzantine-robust aggregation — Krum, Multi-Krum, coordinate-wise trimmed mean and median, and similarity-based trust weighting — replacing plain FedAvg so that a bounded fraction of adversarial updates cannot dominate the global model. Statistical heterogeneity across sites is handled through FedProx-style proximal regularisation and FedBN-style local batch normalisation.

The framework is instantiated on multi-institutional chest radiograph data partitioned across simulated hospital nodes, performing multi-label pathology detection and ordinal severity grading. Evaluation compares the federated global model against isolated single-site baselines and a centralised upper bound, and the system is audited empirically — gradient-inversion and membership-inference attacks are executed against the shared updates to verify that the claimed privacy is actually realised, rather than assumed by construction. As of this submission, the complete framework — the federated core, both defence layers, and the automated test suite — has been implemented and validated on synthetic data; the joint privacy–robustness composition study and full-scale results on real data are in progress.

Keywords — *Federated Learning, Differential Privacy, Byzantine-Robust Aggregation, Medical Diagnosis, Chest Radiography, Privacy Audit.*

TABLE OF CONTENTS

1. INTRODUCTION....4
1.1 Background....4
1.2 Motivation....4
1.3 Scope of the Project....4
2. PROJECT DESCRIPTION AND GOALS....5
2.1 Literature Review....5
2.2 Research Gap....9
2.3 Objectives....9
2.4 Problem Statement....10
2.5 Project Plan....11
2.5.1 Current Progress (as of Review 2)....11
3. TECHNICAL SPECIFICATION....12
3.1 Requirements....13
3.1.1 Functional....13
3.1.2 Non-Functional....13
3.2 Feasibility Study....14
3.2.1 Technical Feasibility....14
3.2.2 Economic Feasibility....14
3.2.3 Social Feasibility....14
3.3 System Specification....15
3.3.1 Hardware Specification....15
3.3.2 Software Specification....15
4. DESIGN APPROACH AND DETAILS....15
4.1 System Architecture....15
4.2 Design....16
4.2.1 Data Flow Diagram....16
4.2.2 Use Case Diagram....17
5. REFERENCES....18

1. INTRODUCTION

1.1 Background

Hospitals generate large volumes of medical imaging data, yet regulatory frameworks such as HIPAA, the GDPR, and India's Digital Personal Data Protection (DPDP) Act prevent this data from being centralised for machine learning. Individual healthcare institutions are consequently left training diagnostic models on small, demographically narrow, and acquisition-biased local datasets, which limits the accuracy and generalisability of the resulting systems. Federated learning (FL) has emerged as a way around this constraint: instead of moving patient data, only model updates are exchanged with a coordinating server, which aggregates them into a shared global model. However, FL by itself provides no formal privacy guarantee — exchanged gradients can still be inverted to reconstruct training images or to determine whether a particular patient's record was used in training. FL's open participation model also creates a new vulnerability, since any hospital node whose infrastructure is compromised, or which experiences a data-quality fault, can contribute a corrupted update capable of poisoning the shared model. This project is set against that backdrop, using multi-institutional chest radiograph diagnosis as the concrete clinical task through which the underlying privacy and robustness problem is designed, implemented, and evaluated.

1.2 Motivation

The motivation for this project follows directly from a gap between the federated learning literature and what a real hospital deployment would need. Large-scale clinical FL studies have shown that collaboration measurably improves diagnostic accuracy and generalisability over training at a single site, but these deployments typically use plain federated averaging with no formal privacy budget and no defence against faulty or malicious participants. Conversely, work that provides a strong theoretical privacy guarantee, or a strong Byzantine-robust aggregation rule, is almost always evaluated in isolation from the other — differential privacy is tested assuming every client is honest, and robust aggregation is tested assuming every update is noise-free. In practice, a hospital-grade system needs both guarantees simultaneously, and the mechanisms behind them are already known to conflict: the very noise that protects patient privacy also destroys the statistical signal that a robust aggregator relies on to detect a malicious update. This project is motivated by the need to study that interaction directly, rather than assume two individually sound defences remain sound when deployed together, and to validate the resulting design on a clinically meaningful task rather than a synthetic benchmark.

1.3 Scope of the Project

This project covers the design, implementation, and empirical evaluation of a hierarchical federated learning framework for multi-institutional chest radiograph diagnosis, simulating three hospital nodes coordinated by a central aggregation server. The scope includes a configurable data pipeline for the NIH ChestX-ray14 dataset, including a heuristic severity-label derivation for ordinal grading; a dual-head diagnostic model performing multi-label

pathology detection and per-disease severity grading; a Flower-based federated training loop supporting both IID and non-IID data partitions; a client-level differential privacy layer with formal privacy-budget accounting; five Byzantine-robust aggregation strategies together with four adversarial client behaviours used to test them; and an experimental framework for jointly sweeping privacy budget, attacker fraction, and heterogeneity level. The scope explicitly excludes deployment in a live clinical environment, integration with real hospital IT infrastructure, and regulatory certification — the project is a research and demonstration testbed, not a production clinical system. A FastAPI backend and a React-based dashboard for visual demonstration of the federated process are included as a supporting deliverable rather than the primary research contribution.

2. PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

This section reviews ten papers most directly relevant to the project's two central mechanisms — differential privacy and Byzantine-robust aggregation — in federated medical learning, drawn from work published between 2021 and 2026.

[1] Federated Learning for Thoracic Disease Classification Using Convolutional Neural Networks and Differential Privacy

2026 — Healthcare Technology Letters

Proposes a federated framework with differential privacy for multi-label thoracic disease classification from chest radiographs, evaluated on NIH ChestX-ray14 and Stanford CheXpert. It explicitly targets resource-constrained clinical environments where data sharing is legally restricted, and studies the cost of adding DP on top of FL.

- Directly quantifies the privacy–utility penalty of DP integration — both accuracy degradation and computational overhead — in a realistic multi-label radiology setting.
- Establishes CNN-based DP-FL as a workable, reproducible baseline for chest X-ray diagnosis under decentralised training.
- Limitation relevant to this work: all participating clients are assumed honest. No adversarial/Byzantine threat model, no robust aggregation, and no empirical privacy audit against reconstruction attacks — the privacy claim rests entirely on the analytic ε.

[2] OptiGradTrust: Byzantine-Robust Federated Learning with Multi-Feature Gradient Analysis and Reinforcement Learning-Based Trust Weighting

2025

A defence framework for federated medical imaging that scores every client update using a six-dimensional gradient “fingerprint” (VAE reconstruction error, cosine-similarity metrics, L2 norm, sign-consistency ratio, Monte-Carlo Shapley value), fed into a hybrid reinforcement-learning/attention module producing adaptive trust weights before aggregation. Also

introduces FedBN-Prox, combining federated batch normalisation with proximal regularisation for stable convergence under heterogeneity.

- Evaluated on MNIST, CIFAR-10, and an Alzheimer's MRI dataset against five distinct Byzantine attack types, under both IID and non-IID splits, outperforming established defences including FLTrust and FLAME.
- Treats heterogeneity and adversarial behaviour as coupled problems — the optimizer is co-designed with the defence rather than bolted on.
- Gap: addresses integrity but not confidentiality — updates are aggregated in the clear, with no differential privacy guarantee.

[3] Privacy-Preserving Chest X-ray Report Generation via Multimodal Federated Learning with ViT and GPT-2

2025

A federated multimodal pipeline (Vision Transformer encoder + GPT-2 decoder) that generates radiology reports without centralising images, comparing three aggregation strategies: FedAvg, Krum, and a loss-aware federated averaging variant.

- Demonstrates that a Byzantine-resilient aggregator (Krum) can outperform plain FedAvg on clinical output quality (ROUGE, BLEU, BERTScore, RaTEScore) even in a benign federation.
- Shows federated training matching or surpassing centralised training, countering the assumption that decentralisation necessarily costs accuracy.
- Establishes robust aggregation as a first-class design choice in medical FL rather than a pure security add-on. No DP layer is applied.

[4] Fusion of Personalized Federated Learning (PFL) with Differential Privacy (DP) Learning for Diagnosis of Arrhythmia Disease

2025 — PLOS ONE

A privacy-preserving architecture fusing personalised federated models with local differential privacy for 12-lead ECG arrhythmia diagnosis, using GRU-based temporal modelling and feature fusion across decentralised institutions.

- Reports approximately 93.7% average accuracy over a 10,646-recording ECG corpus while enforcing local DP, outperforming both centralised and federated baselines.
- Evidence that the FL + DP + personalisation stack is modality-agnostic, generalising beyond imaging to signal-based diagnostics — supporting the framing of federated clinical AI as an architecture-level contribution rather than a task-specific one.

[5] Efficient Differential Privacy Enabled Federated Learning Model for Detecting COVID-19 Disease Using Chest X-ray Images

2024 — Frontiers in Medicine

Introduces an adaptive DP-FL model in which the privacy level is adjusted dynamically according to real-time data-sensitivity analysis, coupled with an early-stopping mechanism that limits privacy-budget consumption during training.

- Moves beyond a single fixed ε: adaptive noise calibration improves the practical utility of DP-FL across diverse hospital environments.
- Explicitly analyses the joint effect of early stopping on both utility and privacy, i.e. budget spend as a controllable training hyperparameter.
- Motivates adaptive rather than static budget allocation in the proposed system.

[6] Federated Learning Enables Big Data for Rare Cancer Boundary Detection

2022 — Nature Communications

The largest federated study in medicine at the time of publication: 71 sites across 6 continents collaboratively training a glioblastoma tumour-boundary detector (n = 6,314) by sharing only numerical model updates.

- Reports a 33% delineation improvement for the surgically targetable tumour and 23% for the full tumour extent over a publicly trained model.
- Strongest existing evidence that federated aggregation genuinely improves out-of-sample generalisability for rare diseases and underrepresented populations — the core value proposition of this project.
- Gap: no formal DP guarantee and no adversarial threat model; mutual trust between the 71 sites is assumed by construction.

[7] Defending Against Reconstruction Attacks Through Differentially Private Federated Learning for Classification of Heterogeneous Chest X-Ray Data

2022 — Sensors

Evaluates DP-FL specifically as a defence against privacy attacks, comparing DenseNet121 and ResNet50 across an imbalanced, heterogeneous federation of 36 clients constructed from CheXpert and Mendeley chest X-ray data.

- Non-private baselines reach AUC ≈ 0.94, but the authors demonstrate that both architectures leak: image-reconstruction attacks successfully recover training data from individual client updates.
- Identifies DenseNet121 as the more DP-friendly backbone, i.e. architecture choice interacts with the privacy mechanism.
- The clearest published evidence that FL without DP is not privacy-preserving — the direct motivation for this project's empirical privacy-audit component.

[8] Federated Learning and Differential Privacy for Medical Image Analysis

2022 — Scientific Reports

A systematic case study of DP-FL on histopathology whole-slide images from The Cancer Genome Atlas, varying the number of participating healthcare providers, per-site dataset size, and IID vs. non-IID distribution.

- Isolates federation topology as an independent experimental variable, showing that client count and data skew — not only the privacy budget — determine achievable accuracy.
- Provides the experimental-design template for the heterogeneity and client-count sweeps proposed here.
- Confirms that non-IID skew compounds the DP utility penalty, rather than acting independently of it.

[9] End-to-End Privacy Preserving Deep Learning on Multi-Institutional Medical Imaging

2021 — Nature Machine Intelligence

Presents PriMIA, an open-source framework combining differentially private federated training, secure aggregation, and encrypted remote inference, validated on expert-level paediatric chest X-ray classification.

- Achieves classification performance on par with locally, non-securely trained models while providing both theoretical and empirical privacy guarantees.
- Demonstrates that the deployed protections prevent recovery of usable data by a gradient-based model-inversion attack — one of the few works to verify, not merely assert, its privacy claim.
- Closest existing precedent for an end-to-end secured medical FL stack; however, it models a curious server/adversary only, not malicious participants, so integrity under Byzantine clients is out of scope.

[10] Federated Learning for Predicting Clinical Outcomes in Patients with COVID-19

2021 — Nature Medicine

The EXAM study: a federated model trained across 20 institutes worldwide on vital signs, laboratory values, and chest X-rays, predicting future oxygen requirements of symptomatic COVID-19 patients.

- Achieved average AUC > 0.92 for outcomes at 24 h and 72 h from emergency-room presentation.
- Reported a 16% average AUC improvement and a 38% average increase in generalisability over models trained at a single site on that site's own data — the canonical quantitative justification for this project's core hypothesis that collaboration outperforms isolation.
- Showed FL enabling rapid multi-national collaboration across heterogeneous, unharmonised datasets without data exchange.

- Gap: standard federated averaging, no declared privacy budget, no robustness mechanism.

2.2 Research Gap

The literature reviewed above converges on six specific, unresolved gaps that this project directly addresses:

- **G1 — Privacy and robustness are studied in isolation, never jointly.** DP-FL work ([1], [5], [7], [8], [9]) assumes an honest-but-curious server with fully cooperative clients. Byzantine-robust work ([2], [3]) assumes plaintext, noise-free updates. No study systematically characterises their interaction, despite a direct conflict: DP gradient clipping and Gaussian noise deliberately destroy the very statistical structure (update norm, direction, inter-client similarity) that robust aggregators such as Krum, trimmed-mean, and FLTrust rely on to identify malicious contributions. A defence that is sound in isolation may fail under composition.
- **G2 — Clinically validated deployments carry no formal guarantees.** The large-scale studies that prove FL works clinically ([6], [10]) use plain FedAvg with an implicit trust assumption, no stated ε, and no adversarial threat model. Conversely, frameworks with strong formal guarantees ([9]) are demonstrated only at small scale. There is no bridge between clinical credibility and cryptographic/statistical rigour.
- **G3 — Privacy is asserted rather than audited.** Most medical FL literature treats privacy preservation as an architectural property that follows automatically from not moving data. Only a minority ([7], [9]) execute reconstruction attacks against their own system. The field lacks the standard practice of reporting measured residual leakage at a declared privacy budget.
- **G4 — Heterogeneity interacts with both defences, and is unquantified.** Non-IID skew arising from scanner vendors, acquisition protocols, and differing disease prevalence is handled by FedProx/FedBN-style methods ([2], [8]), but their behaviour under simultaneous DP clipping and robust aggregation is unstudied. Honest minority-distribution clients can be statistically indistinguishable from attackers, so a robust aggregator risks silently excluding exactly the underrepresented sites that federation was designed to include — a fairness failure with direct clinical consequences.
- **G5 — Diagnostic granularity is limited to presence/absence.** Federated radiology work is dominated by binary or multi-label pathology detection. Ordinal severity grading, which is what actually drives triage and clinical decision-making, remains essentially unexplored under federated and privacy-constrained training.
- **G6 — No unified, reproducible testbed.** Results across the literature are not comparable, because each work fixes a different subset of {privacy budget, attacker fraction, heterogeneity level, aggregation rule, backbone}. No open framework exists that allows these axes to be swept jointly on a common medical benchmark.

2.3 Objectives

Guided by the gaps identified above, the project defines the following objectives:

1. **Design a hierarchical federated architecture** for multi-institutional medical diagnosis — a coordinating centre node with N leaf nodes — exposing a modular interface so that the aggregation rule and the privacy mechanism can be swapped independently.
2. **Integrate client-level differential privacy** into local training through per-sample gradient clipping and calibrated Gaussian noise, with a Rényi-DP accountant reporting a formal cumulative (ε, δ) budget for the complete training run.
3. **Implement and compare Byzantine fault-tolerant aggregation rules** — Krum/Multi-Krum, coordinate-wise trimmed mean and median, and similarity-based trust weighting — against FedAvg under label-flipping, sign-flipping, scaled-update, and model-replacement attacks at varying adversarial client fractions.
4. **Characterise the joint privacy–robustness–utility trade-off** as the primary research contribution: measure diagnostic performance as a function of (ε, attacker fraction, heterogeneity level) and identify the regimes in which DP noise measurably degrades the detection capability of robust aggregation.
5. **Mitigate statistical heterogeneity** across simulated hospital nodes using proximal regularisation and local batch normalisation, and verify that robust aggregation does not systematically exclude honest clients holding minority data distributions (fairness-under-robustness check).
6. **Validate on chest radiography** with two prediction heads — multi-label pathology detection and per-disease ordinal severity grading (mild / moderate / severe) — benchmarking the federated global model against isolated single-site models and a centralised upper bound.
7. **Empirically audit the privacy claim** by executing gradient-inversion and membership-inference attacks against intercepted client updates, reporting measured reconstruction quality (SSIM/PSNR) and attack success rate at each privacy budget, instead of asserting privacy by construction.
8. **Release a reproducible, configuration-driven testbed** and reference implementation that allows the privacy, robustness, and heterogeneity axes to be varied independently, enabling comparable evaluation of future federated clinical AI systems.

2.4 Problem Statement

Existing federated learning systems for medical diagnosis treat differential privacy and Byzantine-robust aggregation as independent defences, each validated under the assumption that the other threat does not exist: differential privacy is evaluated assuming a fully honest set of clients, while Byzantine-robust aggregation is evaluated assuming noise-free, unprotected updates. Because DP's gradient clipping and calibrated noise directly alter the statistical properties — norm, direction, and inter-client similarity — that robust aggregators use to

distinguish honest updates from malicious ones, a system that deploys both defences together cannot be assumed to inherit the guarantees each provides individually. At the same time, published clinical-scale federated learning deployments demonstrate real diagnostic benefit but carry no formal privacy budget and no threat model, while systems with strong formal guarantees remain unvalidated at clinical scale or under adversarial conditions. This project addresses the specific, unresolved problem of characterising whether — and under what combinations of privacy budget, adversarial client fraction, and data heterogeneity — client-level differential privacy and Byzantine-robust aggregation can be composed without one undermining the other, using multi-institutional chest radiograph diagnosis as the validating clinical task.

2.5 Project Plan

The project timeline below organises development into sixteen tasks, from environment setup through to the final report and presentation, against which progress is tracked across both project reviews.

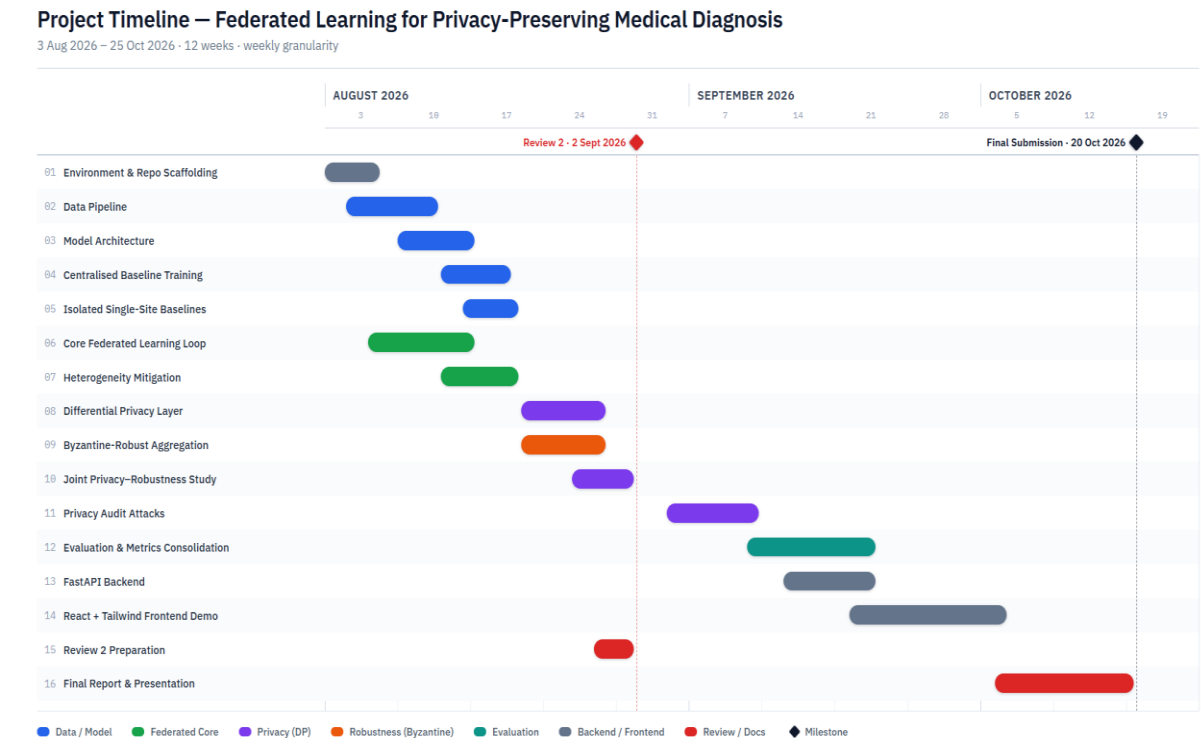


Fig. 1. Project timeline (Gantt chart)

2.5.1 Current Progress (as of Review 2)

Tasks 1 through 9 of the project plan have been implemented and are functioning end-to-end:

- Environment and project setup, with a Python 3.11 environment, all required libraries installed, and the repository structure and documentation in place.

- A data pipeline that loads the chest radiograph labels, derives the mild / moderate / severe severity labels, and partitions the data across three simulated hospital nodes under both even and uneven (non-IID) splits.
- A dual-head DenseNet/ResNet-based model producing both disease-detection and severity-grading outputs.
- Centralised and isolated-site baseline training, representing the best-case (pooled data) and worst-case (single hospital) performance references.
- The core federated learning demonstration: three simulated hospitals train locally and a central node aggregates their models via FedAvg over repeated rounds — the project's main deliverable, functional end-to-end.
- Heterogeneity handling via FedProx and FedBN, so the system copes with hospitals holding very different data distributions.
- A differential-privacy layer that protects patient data during training and reports the cumulative privacy budget (ε).
- Byzantine attack simulation and five robust aggregation rules capable of surviving those attacks.

All of the above is covered by six automated test suites, all currently passing, which run on small synthetic data (no dataset download required) and complete in under a minute. Supporting documentation — a setup guide, a running progress tracker, and a research log — is maintained alongside the code.

The work remaining before the final review is:

- The main research experiment (Task 10) — running the differential-privacy and attack-defence mechanisms together and sweeping the combinations to measure how they compose. The individual components already exist; what remains is the sweep script itself.
- The privacy audit (Task 11) — executing gradient-inversion and membership-inference attacks to empirically confirm that the stated privacy budget actually protects patient data, rather than assuming that it does.
- Full-scale results on real data — all current results use small synthetic data for testing; the real numbers require a GPU training run on the complete NIH ChestX-ray14 dataset, planned on Google Colab.
- The FastAPI backend and React dashboard for a live visual demonstration of the federated process.

In summary, the complete system has been built and tested end-to-end; what remains before the final submission is running the full-scale experiments on real data and building the demonstration interface.

3. TECHNICAL SPECIFICATION

3.1 Requirements

3.1.1 Functional

- **FR1.** The system shall train the model locally on each hospital node without transmitting raw patient data off-site.
- **FR2.** The system shall apply per-sample gradient clipping and calibrated Gaussian noise to every client update before transmission (client-level DP-SGD).
- **FR3.** The central server shall support pluggable aggregation strategies — FedAvg, Krum/Multi-Krum, trimmed mean, median, and trust weighting — selectable via configuration.
- **FR4.** The system shall simulate Byzantine/malicious client behaviour (label-flip, sign-flip, scaled-update, model-replacement) for a configurable fraction of nodes.
- **FR5.** The system shall track and report the cumulative differential privacy budget (ε, δ) for every training run via a Rényi-DP accountant.
- **FR6.** The evaluation module shall report multi-label pathology detection and per-disease severity grading performance for the global model after each aggregation round.
- **FR7.** The system shall be able to execute gradient-inversion and membership-inference attacks against intercepted client updates and report the resulting reconstruction quality and attack success rate.
- **FR8.** The system shall expose a FastAPI backend and a React + Tailwind CSS dashboard to visualise per-node training status, weight-update flow, and aggregation events.

3.1.2 Non-Functional

- **Privacy:** The system shall enforce a formally accounted differential-privacy budget (ε, δ) on every client update, rather than relying on architectural assumptions alone.
- **Robustness:** The system shall tolerate a bounded fraction of Byzantine or malicious client updates without a proportionate collapse in global model accuracy.
- **Performance:** Local client training rounds shall complete within a practical time budget on commodity GPU hardware, including free-tier cloud GPU runtimes, to keep experimentation iterations short.
- **Scalability:** The federated core shall be agnostic to the number of client nodes, so the same codebase can be extended from three nodes to a larger simulated federation without structural changes.
- **Reliability:** Automated test suites shall validate every module — data, model, federated loop, privacy, and robustness — against small synthetic data, so regressions are caught without requiring a full dataset download.

- **Reproducibility:** Every experiment configuration — privacy budget, attacker fraction, heterogeneity level, and aggregation rule — shall be fully specified in a versioned configuration file so results can be regenerated exactly.
- **Maintainability:** Aggregation strategies and privacy mechanisms shall be pluggable components behind a common interface, so new strategies can be added without modifying the core training loop.
- **Usability:** The demonstration dashboard shall present per-node training status and aggregation events in a form understandable to a non-specialist reviewer.

3.2 Feasibility Study

3.2.1 Technical Feasibility

The project relies entirely on established, actively maintained open-source technologies — PyTorch for model training, Flower for federated orchestration, per-sample gradient clipping and noise injection for differential privacy, and FastAPI/React for the demonstration layer — all of which are well documented and widely used in both academic and applied federated learning research. The core federated simulation, differential-privacy layer, and all five Byzantine-robust aggregation strategies have already been implemented and validated against automated test suites on synthetic data, which substantially de-risks the remaining work. The main technical dependency still outstanding is access to GPU compute for full-scale training on the NIH ChestX-ray14 dataset, which is addressed using Google Colab's GPU runtime rather than requiring dedicated institutional infrastructure.

3.2.2 Economic Feasibility

As an academic research project, the system is built entirely on free and open-source software, and the computational requirements for full-scale training are met using free-tier or low-cost cloud GPU access (Google Colab) rather than dedicated hardware procurement. There is no licensing cost, no proprietary dataset cost (NIH ChestX-ray14 is publicly released for research use), and no ongoing infrastructure cost beyond incidental cloud compute during the training and sweep phases, making the project economically feasible within a standard academic project budget.

3.2.3 Social Feasibility

The project addresses a socially significant constraint on clinical AI adoption — the inability of hospitals to pool patient data for model training — and its output is a research testbed rather than a deployed clinical tool, so it does not raise the direct patient-facing risks that would accompany a live diagnostic system. The chest radiograph dataset used is a publicly available, de-identified research dataset, avoiding direct patient-privacy exposure during development. The broader ethical consideration the project engages with directly is the reliability of privacy claims made about federated medical AI systems: rather than assuming that a stated privacy budget is meaningful, the project's own audit component is designed to verify that claim

empirically, which aligns with responsible-AI expectations around verifiable rather than assumed privacy protection.

3.3 System Specification

3.3.1 Hardware Specification

- A standard development machine (multi-core CPU, 8 GB or more RAM) for code development, testing, and small-scale/synthetic-data validation.
- A CUDA-capable GPU, accessed via Google Colab's GPU runtime, for full-scale model training and the experimental sweep on the complete NIH ChestX-ray14 dataset.
- Sufficient local and cloud storage (on the order of tens of gigabytes) to hold the chest radiograph dataset and its partitioned, per-node copies.

3.3.2 Software Specification

- **Operating System:** Windows / Linux / macOS for development; Linux for the Google Colab runtime.
- **Programming Language:** Python 3.11.
- **Development Environment:** Visual Studio Code; Jupyter and Google Colab notebooks for GPU training runs.
- **Libraries and Frameworks:** PyTorch, Flower (flwr), differential-privacy utilities (per-sample gradient clipping and Rényi-DP accounting), scikit-learn, NumPy, pandas, FastAPI, React, and Tailwind CSS.
- **Version Control:** Git and GitHub.
- **Testing:** pytest-based automated test suites.
- **Data / Storage:** Local filesystem and CSV-based experiment logs; no dedicated database is required.

4. DESIGN APPROACH AND DETAILS

4.1 System Architecture

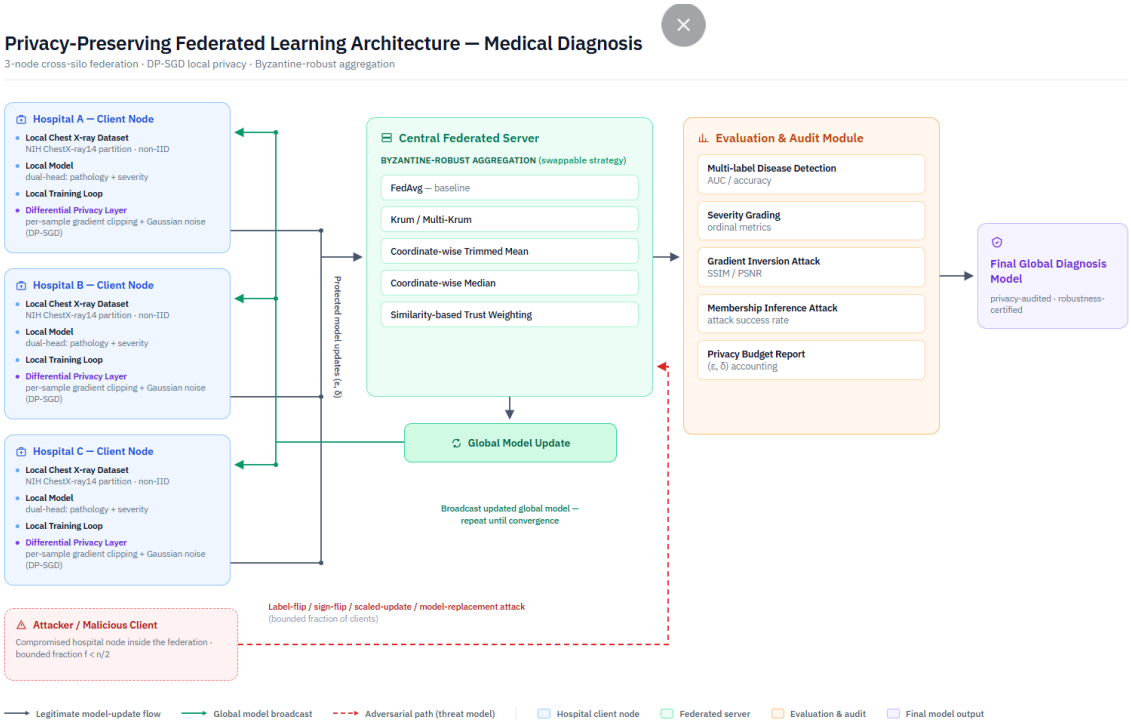


Fig. 2. System architecture

Figure 2 shows the system's hierarchical client–server federated architecture. Three simulated hospital nodes each hold a private, non-identically-distributed partition of the chest radiograph dataset and train the shared dual-head model locally; before any update leaves a node, per-sample gradient clipping and calibrated Gaussian noise are applied under a Rényi-DP accountant that tracks the cumulative privacy budget. The central federated server receives these protected updates and aggregates them using one of several interchangeable Byzantine-robust strategies — FedAvg as the baseline, or Krum, Multi-Krum, coordinate-wise trimmed mean, coordinate-wise median, or similarity-based trust weighting — so that a bounded fraction of adversarial or faulty client updates cannot dominate the resulting global model. The updated global model is broadcast back to all nodes and the process repeats until convergence, after which the evaluation and audit module measures diagnostic performance and executes gradient-inversion and membership-inference attacks against the exchanged updates to verify the claimed privacy protection empirically.

4.2 Design

4.2.1 Data Flow Diagram

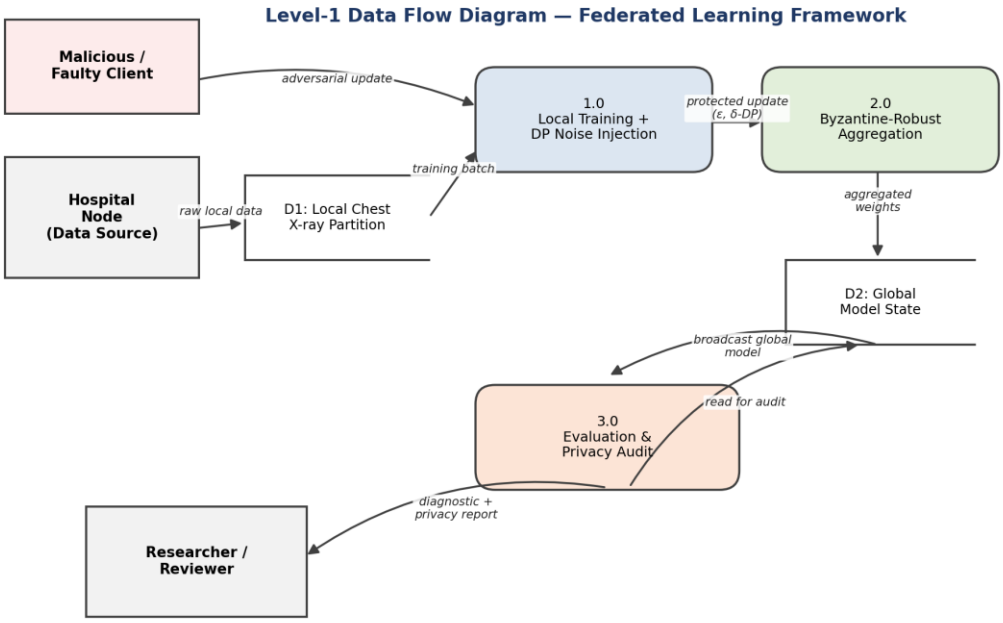


Fig. 3. Level-1 data flow diagram

The data flow diagram traces information through the system in three stages. In the first stage, each hospital node's local chest X-ray partition is used to train the local model, with per-sample gradient clipping and Gaussian noise applied before the resulting update leaves the node; a malicious or faulty client may instead inject an adversarial update at this same stage. In the second stage, the central server aggregates all received updates using the configured Byzantine-robust strategy and writes the result to the global model state, which is broadcast back for the next round. In the third stage, the evaluation and privacy-audit module reads the global model state to measure diagnostic performance and to execute gradient-inversion and membership-inference attacks, producing a diagnostic and privacy report for the researcher or reviewer.

4.2.2 Use Case Diagram

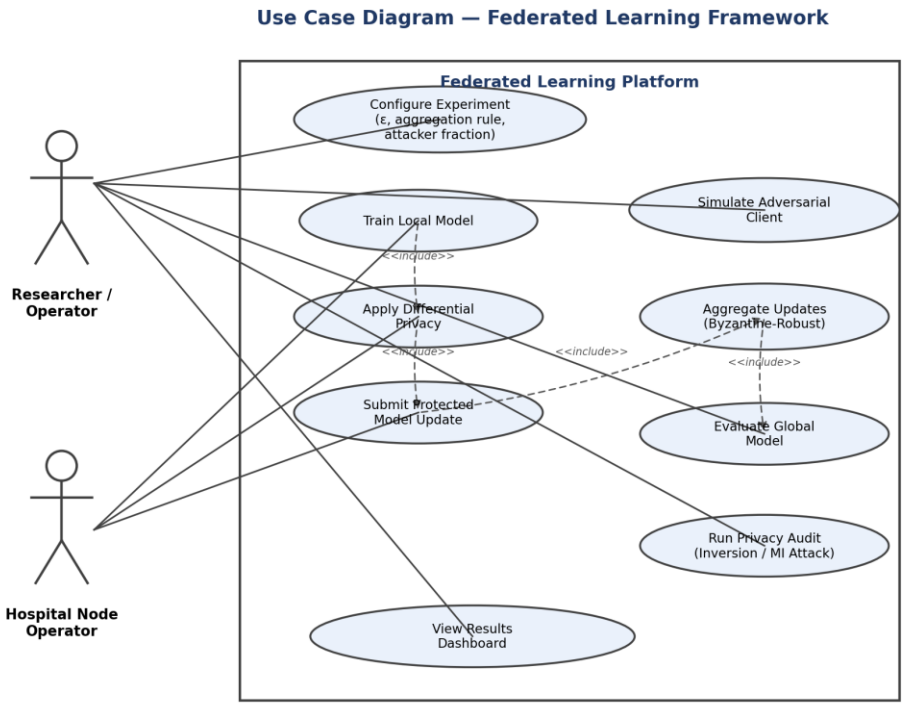


Fig. 4. Use case diagram

Two actors interact with the platform. The hospital node operator trains the local model, applies differential privacy to the resulting update, and submits the protected update to the central server. The researcher or operator configures each experiment (privacy budget, aggregation rule, and attacker fraction), can simulate an adversarial client to test the system's defences, triggers aggregation and evaluation of the global model, runs the privacy audit, and views the results through the dashboard. The include relationships reflect that training a local model necessarily invokes the differential-privacy step, which in turn produces the update that is submitted and aggregated, and that aggregation is followed by evaluation of the resulting global model.

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